Layered Feature Story

Repository URL:

https://github.com/glebans-Lebans/ASSESSMENT-Layered-Feature-Story-Starter

Published Site URL:

https://glebans-lebans.github.io/ASSESSMENT-Layered-Feature-Story-Starter/

Test Record

HTML validation

Document checking completed. No errors or warnings to show.

Outline

- <h1> AZ Water Association's Water Reuse Project of The Year
  - <h2> Fountain Park ASR Well Vault Replacements
  - <h2> In this story
  - <h2> L2-L3 · Flow and Positioned Relationships
    - <h3> Original or Innovative Application of New or Existing Techniques
    - <h3> Social and Economic Considerations
    - <h3> Complexity
  - <h2> L4-L5 · Layers, Shapes, and Fallback
    - <h3> Meeting and Exceeding Owner Needs
  - <h2> Contributors
  - <h2> L6 · Documentation and Testing

Used the HTML parser.

Total execution time 9 milliseconds.

- I validated using the W3C Nu HTML validator and received no errors.

CSS validation

W3C CSS Validator results for styles.css (CSS level 3 + SVG)

Sorry! We found the following errors (3)

URI : styles.css

195		Property `shape-outside` doesn't exist : `circle(50%)`
198	`.shape-portrait`	Property `shape-outside` doesn't exist : `circle(50%)`
199	`.shape-portrait`	Property `shape-margin` doesn't exist : `1rem`

↑ TOP

Interested in "developing" your developer skills? In W3Cx's hands-on Professional Certificate Program, learn how to code the right way by creating Web sites and apps that use the latest Web standards. Find out more!

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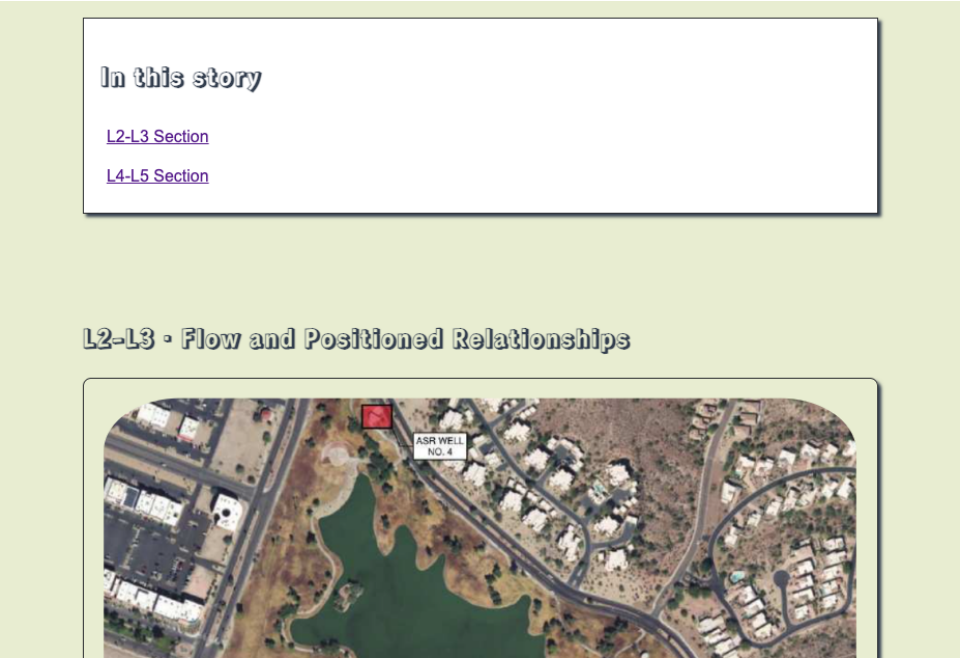
- I validated my styles.css with W3C CSS Validator and got errors saying that shape-outside and shape-margin properties do not exist. These properties do exist and are listed on the MDN website with full support in the 12 most frequently used browsers including Chrome, Edge, Firefox, and Safari.
- MDN shape-outside
    - https://developer.mozilla.org/en-US/docs/Web/CSS/Reference/Properties/shape-outside
- MDN shape-margin
    - https://developer.mozilla.org/en-US/docs/Web/CSS/Reference/Properties/shape-margin

CSS-disabled reading order

CSS rules to alter reading order

Standard reading order



- Grid rules create three columns to change the reading order for the two navigation elements. The “In this story” article navigation is moved from the start of the article content to the left most column. The article content is placed in the second column. The “Contributors navigation is moved to the rightmost column. The navigation sections can be read out of order to the bulk of the article content.

Viewport

320px

- At 320px, images are responsive and shrink to fit viewport width. Float is disabled on circular image and text moves below the image rather than wrapping around it. Sticky elements lose their positioning and sticky functionality and move back in line with the rest of the content. Article navigation is at the start of the main container and Contributors navigation is at the end of the main container.

768px

Quarter Three 2026

L1 · Feature Story

AZ Water Association's Water Reuse Project of The Year

Fountain Park ASR Well Vault Replacements

For more than two decades, Aquifer Storage and Recovery (ASR) wells located throughout Fountain Park have played a critical role in the Fountain Hills Sanitary District's recycled water system. As the underground well vault aged and deteriorated, the District launched a phased infrastructure improvement program to modernize the facilities, improve operational safety, and enhance one of the town's most recognizable public spaces.

In this story

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L4-L5 Section

L2–L3 · Flow and Positioned Relationships

L2–L3 · Flow and Positioned Relationships

ASR Wells at Fountain Lake. Image by AZ Water Association. Source

The District's ASR system supports Fountain Hills' longstanding commitment to water reuse. During periods of low irrigation demand, reclaimed water is stored underground in ASR wells and later recovered during periods of peak demand to irrigate local parks and golf courses. Several of the wells serving Fountain Park and Lake were originally installed more than twenty years ago within underground steel vault structures.

Over time, the steel vaults became severely corroded and exceeded their useful service life. In addition to material deterioration, the below ground configuration required District staff to routinely enter confined spaces to perform daily operation and maintenance activities, creating ongoing safety and accessibility concerns.

To address these challenges, the Fountain Hills Sanitary District partnered with the Town of Fountain Hills, Wilson Engineers, MGC Contractors, Hunter Contracting, and a local architect to implement a phased ASR Well Vault Replacement program across three well sites within Fountain Park. The project included improvements to ASR Wells No. 1, No. 2, and No. 4.

> "The Town, the District, and the general public that use the famous Fountain Park are all winners with this project." - *Fountain Hills Sanitary District Manager Dana Trompke*

Original or Innovative Application of New or Existing Techniques

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ASR Wells No. 2 and No. 4 located on the north and east sides of Fountain Lake, were equipped within new dual use facilities that house critical ASR equipment while also providing much needed public restroom amenities for park visitors. The buildings were designed to complement the surrounding park architecture and blend into the highly visible public setting near the lake, amphitheater, and pedestrian pathways.

Educational signage installed at the facilities highlights the District's recycled water process, illustrating the water cycle from Central Arizona Project delivery and treatment through wastewater collection, aquifer storage, recovery, and irrigation reuse. The displays reinforce the district's long standing "Recycles Every Drop" commitment while helping visitors better understand the importance of water reuse infrastructure in the desert Southwest.

ASR Well No. 1, located along the western side of Fountain Park near Saguaro Boulevard, presented a different set of design challenges. While the site did not require public restroom facilities, the project required careful coordination with pedestrian pathways, existing utilities, flood control infrastructure, and nearby public art features. Six Presidential statues and associated Bill of Rights signage were relocated with approval from the Town's Public Art Commission to improve pedestrian circulation and maintain the character of the area.

The replacement of the underground vault at Well No. 1 with a new above ground control room significantly improved safety and accessibility for District operations staff. The facility was positioned behind an existing ramada to preserve views of the fountain, while sidewalks were realigned to accommodate operational vehicle access without disrupting heavily used pedestrian pathways.

Before and after comparison of well controls at ASR Well No.1. Image by AZ Water Association. Source

Social and Economic Considerations

Because the wells are located throughout one of the Town's busiest public spaces, the project required careful phasing and coordination to maintain uninterrupted ASR functionality and minimize impacts to park operations. Only one of the five ASR wells could be taken out of service at a time, and each well needed to be returned to operation within a two-month period to meet system demands.

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Construction sequencing also had to accommodate major community events held at Fountain Park, including local high school graduation ceremonies near the amphitheater adjacent to Well No. 4. Maintaining safe public access during construction required close coordination with Town staff, adaptive phasing strategies, and ongoing communication with park users.

Complexity

Wilson Engineers, serving as the design engineer and construction administration consultant, applied extensive regional ASR experience to develop flexible construction approaches tailored to each site.

Hunter Contracting and MGC Contractors implemented the phased improvements using a Construction Manager at Risk delivery method that help the District maintain budget and schedule control while minimizing disruptions to park visitors and ongoing operations.

The project team also incorporated several cost saving measures throughout construction. Certain equipment components, including the well pump, instrumentation equipment, and fiber optic systems, were procured directly by the District to support schedule and budget management. Existing downhole flow control valves were reused and tested by District staff prior to start up, reducing replacement costs while maintaining operational reliability. Similar piping systems were implemented at each site to ease the operator functionality.

Beyond the infrastructure improvements themselves, the project demonstrates how utility upgrades can provide meaningful community benefits when paired with thoughtful site planning and public-focused design. New restroom facilities, improve pedestrian circulation, ADA compliant access routes, educational features, and enhanced architectural integration transformed the facilities into community assets rather than purely operational infrastructure.

New ASR Well facility. Image by AZ Water Association. Source

- At 768px, images are responsive and resize to fit viewport width. Float is disabled on circular image and text moves below the image rather than wrapping around it. Sticky elements move back in line with the rest of the content. Contributors navigation has flex-wrap properties for the company logos, so the logos spread to fit the container on multiple lines.

1280px

Quarter Three 2026

L1 · Feature Story

AZ Water Association's Water Reuse Project of The Year

Fountain Park ASR Well Vault Replacements

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L2-L3 · Flow and Positioned Relationships

ASR Wells at Fountain Lake. Image by AZ Water Association. Source

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Contributors

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FOUNTAIN HILLS SANITARY DISTRICT
WILSON ENGINEERS
NGC
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Before and after comparison of well controls at ASR Well No.1. Image by AZ Water Association. Source

Social and Economic Considerations

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New ASR Well facility. Image by AZ Water Association. Source

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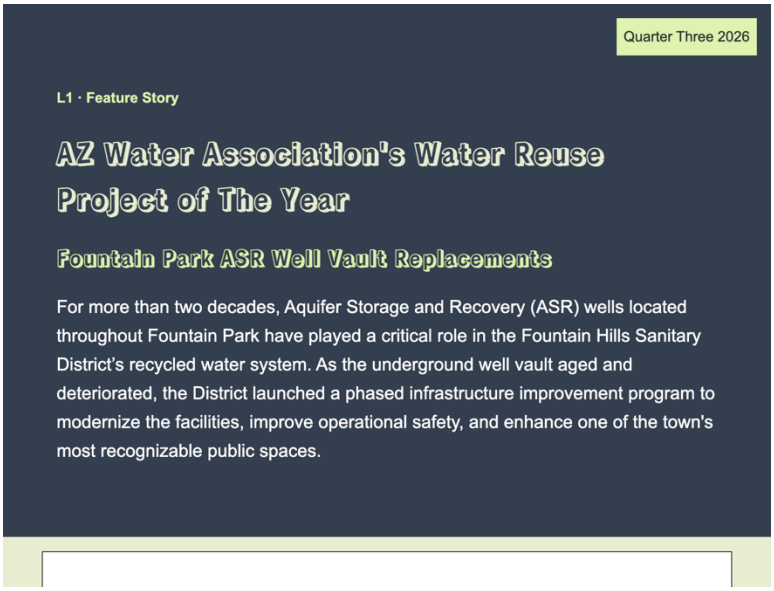
L4-L5 · Layers, Shapes, and Fallback

Meeting and Exceeding Owner Needs

- At 1280px, images are responsive and scale to fit viewport width. Float is enabled on the circular image and text wraps around it in a circular shape. Sticky elements gain grid positioning and sticky functionality. Article navigation is positioned in the first grid column to the left of the article content which is in the second grid column. The Contributors navigation is positioned to the right of the article content in the third grid column.

200% zoom


Quarter Three 2026
L1 · Feature Story
AZ Water Association's Water Reuse Project of The Year
Fountain Park ASR Well Vault Replacements
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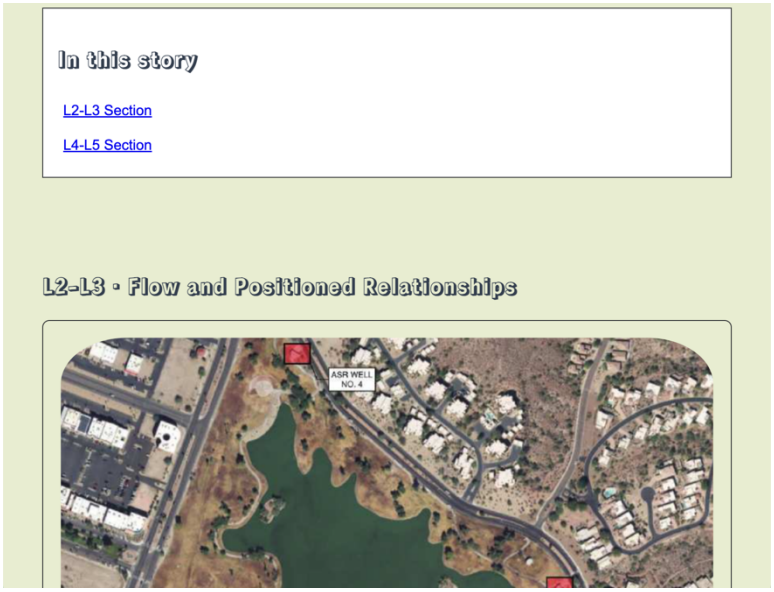
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ASR WELL NO. 4

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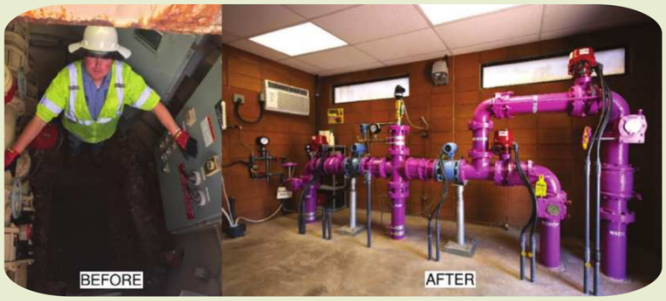

Before and after comparison of well controls at ASR Well No.1. Image by AZ Water Association. Source

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New ASR Well facility. Image by AZ Water Association. Source

L4–L5 · Layers, Shapes, and Fallback

Meeting and Exceeding Owner Needs

According to Fountain Hills Sanitary District Manager Dana Trompke, P.E., the project has been well received by both District staff and the local community.

"The Town, the District, and the general public that use the famous Fountain Park are all winners with this project," Trompke said. "It has been deemed a success by the local community as evidenced by the numerous compliments we receive on the design of the facilities and the appreciation expressed for

Contributors

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FOUNTAIN HILLS SANITARY DISTRICT
Recycles Every Drop

WILSON ENGINEERS

MGC

Hunter

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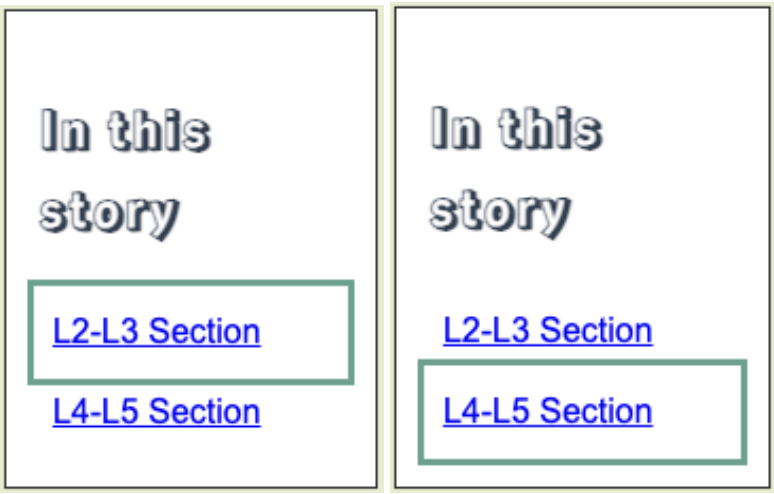
Keyboard Focus



ASR Wells at Fountain Lake. Image by AZ Water Association. Source

Before and after comparison of well controls at ASR Well No.1. Image by AZ Water Association. Source

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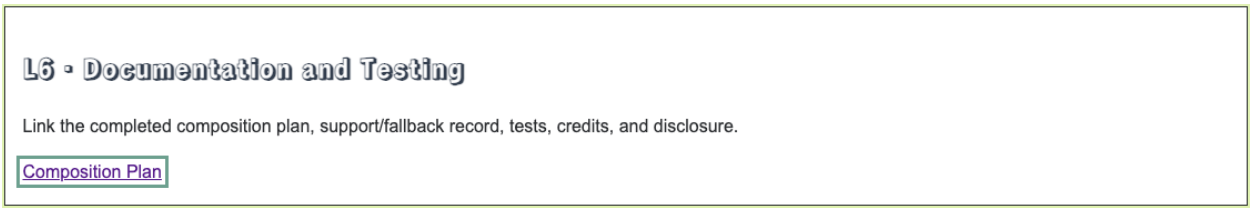

- All links have a focus state with a 0.2rem solid green outline.

Long-title and double-paragraph content growth

- Google drive link for video showing how textual content changes as width is scaled:
  - https://drive.google.com/file/d/1Z1nlOwcD5V37RskOlpve5gkCM9iUpUpi/view?usp=sharing
- Site changes format from single column to three columns as width is adjusted from small to big. Headings and paragraphs wrap text as width is adjusted.

Sticky Behavior

modernize the facilities, improve operational safety, and enhance one of the town's most recognizable public spaces.

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ASR Wells at Fountain Lake. Image by AZ Water Association. Source

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- Article navigation bar on the left side of the page sticks to the top of the page as the user scrolls through the page content. On the right is an external navigation list of contributors for the project in the site content that also sticks to the top of the page as the user scrolls.

Pointer Activation

- Google drive URL for video showing pointer activation on every link:
  - https://drive.google.com/file/d/1C-w77LA-IZP_BU9s7-debGB0ORnTlpVI/view?usp=drive_link

Disabled clip-path

L2-L3 · Flow and Positioned Relationships

ASR Wells at Fountain Lake. Image by AZ Water Association. Source

Before and after comparison of well controls at ASR Well No.1. Image by AZ Water Association. Source

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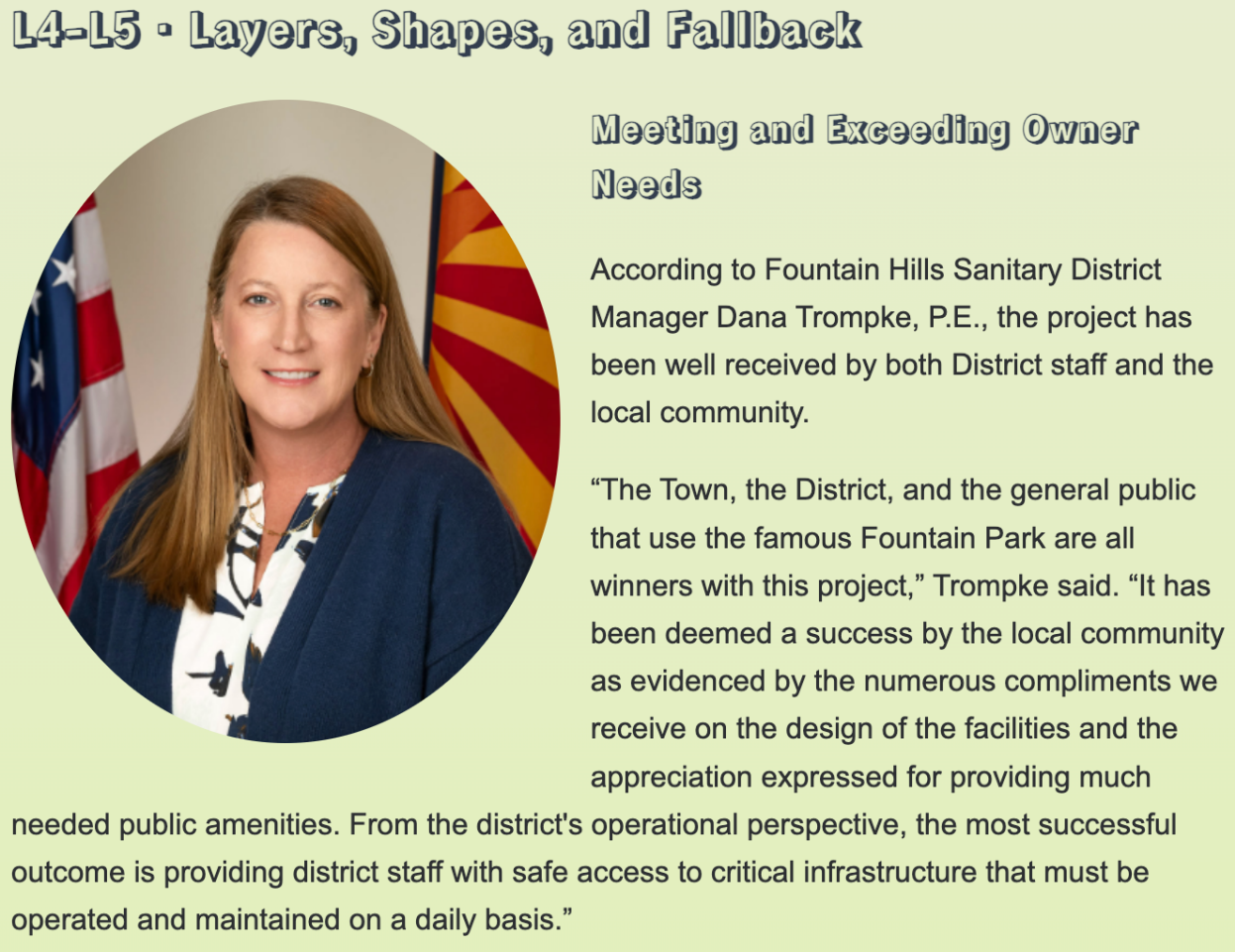
- To test the clip-path for the images, I commented out the rule below.

```
    @supports (clip-path:inset(0.5% round 10%)){
        .feature-figure img{
            clip-path:inset(0.5% round 10%);}
    }
```

- The round edges on the images were removed and replaced with the standard 90-degree angles.

Disabled shape-outside



- To test the shape-outside rule on the floated image, I commented out the line shape-outside:circle(50%) in the feature query below.

```
@supports (shape-outside:circle(50%)){
        .shape-portrait{
            float:inline-start;
            /* shape-outside:circle(50%); */
            shape-margin:1rem;}
        }
    }
```

  - The rounded margin around the image was removed and the text wrapped around the image in a rectangle rather than a circle.

AI Disclosure

No AI used.

Code Defense

I validated my index.html file with the Nu HTML Validator at validator.w3.org/nu and received a warning that my article container was missing a heading such as h1, h2, h3. I checked MDN to make sure this rule needed to be followed. The page for <article> said that headings should be included. The article container was being used as a container for styling, and a heading did not belong within the article, only its sections. I changed the article to a div instead with the class article. After the change, I validated with the Nu HTML Validator once more and received no errors.

References:

All Textual content sourced from:

AZ Water Association

AZ Water News - Newsletter

Quarter Three 2026 | www.azwater.org

https://www.kelmanonline.com/httpdocs/files/AZWater/news3-2026/index.html#

Images:

Picture of Dana Trompke

https://www.az-fhsd.gov/board-of-directors/

Town of Fountain Hills Logo

https://www.fountainhillsaz.gov/

Fountain Hills Sanitary District Logo

https://www.az-fhsd.gov/

Wilson Engineers Logo

https://www.wilson-engineers.com/

MGC Contractors Logo

https://mgccontractors.com/

Hunter Contracting Logo

https://www.huntercontracting.com/